

SECTION 6: TECHNICAL DETAILS FOR BATTERY ENERGY STORAGE SYSTEM

No	Item	Units	Required	Tendered
A	BATTERY MODULE			
	General			
A.1	Manufacturer's Name			Envision
A.2	Country of Origin			China
A.3	Country of Manufacture			China
A.4	Make			BESS
A.5	Model No.			8
A.6	Total Area Required (Acres)			350 m ²
	Dimensions and weight			
A.7	Width/Height/Depth, mm (W/H/D)	mm		6058 (L) x 2438 (W) x 3258 (H) mm
A.8	Weight approximately	kg		< 50000
	Product Compliances and Standards			
A.9	Applicable Standard (latest)		IEC 62619, IEC 62620, IEC 62902, IEC 62485-5, IEC 63056, IEC 62933-1, IEC 62933-2-1, IEC 62933-5-2, UL 9540, UL 9540A, UL 1973 , UN 38.3	UL 9540A, UL 1642, UL 1973, NFPA 69, NFPA 68, NFPA 855 IEC 62933, IEC 62619, IEC 63056, IEC 62477, IEC 61000-6-2, IEC 61000-6-4 UN 38.3, UN3536
A.10	Quality Management		ISO 9001:2015 and ISO 14001:2015	ISO 9001:2015 and ISO 14001:2015
A.11	Place of Testing			China
	BATTERY MODULE DATA			
A.12	No. of Battery Modules			8
A.13	Unit Battery Capacity (P_{rated})	MW		1.5
A.14	Unit Battery Storage Capacity (E_{rated})	MWh		6
A.15	Total Battery Capacity (P_{rated})	MW		12
A.16	Total Battery Storage Capacity (E_{rated})	MWh		48
	BATTERY Module Characteristics			
A.17	Maximum continuous discharge/charge rate (C-rate)		0.25C continuous (compatible with	0.25C continuous

A.18	Depth of Discharge (DoD) limits		4-hour duty)	100%
			≥ 85% usable across operating temperature range	
A.19	Equivalent Full Cycle life (EFCs) at 100% DoD			8000
A.20	Equivalent Full Cycle life (EFCs) at 80% DoD			8800
A.21	Equivalent Full Cycle life (EFCs) at 50% DoD			10000
A.22	Self-Discharge Rate	% / month	≤ 3	≤ 3%
A.23	Thermal management system			Liquid cooling
A.24	Cell, module, and rack configuration			416S1P*6
A.25	Operational Temperature Range	°C		-25°C-45°C

A.26	High voltage safety measures for fire, electrical hazards and automatic shutdown at a fault			yes
A.28	Noise Levels	dB(A)	≤ 80	≤ 80
B	INVERTER / PCS			
	General			
B.1	Manufactures Name			Envision
B.2	Country of Origin			China
B.3	Country of Manufacture			China
B.4	Make			PCS
B.5	Model No.			ENPCS2520
B.6	No. of Inverters			8
B.7	Rated output power	MW,AC		2.52
B.8	Nominal output voltage ($U_{N(AC)}$)	V		690
B.9	Output frequency	Hz		50
B.10	Harmonic distortion, current at rated power	%	≤ 3 (current TDD), ≤ 3 (voltage), per IEEE 519-2022 limits	Meets IEEE 519
B.11	Power factor		0.95 leading and Lagging over full operating envelope	-0.95-0.95
B.12	Droop Settings		1% - 9%	1% - 9%
B.13	Adjustable Frequency Deadband Range		0.0 – 1.0 (settable in 0.05 Hz steps) around 50 Hz	0.0 – 1.0 (settable in 0.05 Hz steps) around 50 Hz
	Dimensions and weight			

B.14	Width/Height/Depth, mm (W/H/D)	mm		12192 mm x 2438 mm x 2896 mm (For the whole 10MVA MV station)
B.15	Weight approximately	kg		49000 (For the whole 10MVA MV station)
	Environmental Limits			
B.16	Degree of protection			IP65
B.17	Ambient temp. range (nominal ratings)	°C		-25°C-45°C
B.18	Relative humidity, not condensing			100%, non-condensation
B.19	Maximum noise level			≤85dB(A) @1m
B.20	DC disconnection device			yes
B.21	AC disconnection device			yes
B.22	DC overvoltage protection (surge protection)			yes
B.23	AC overvoltage protection (surge protection)			yes
	Protection			
B.24	Ground fault monitoring		Yes	yes
B.25	Grid monitoring		Yes	yes
B.26	frequency		Yes	yes
B.27	voltage		Yes	yes
B.28	Anti-islanding		Yes	yes
B.29	DC reverse polarity			yes
B.30	AC and DC short circuit and over current			yes
B.31	AC and DC over voltage and temperature			yes
	User interface and communications			
B.32	Local user interface		Yes	yes
B.33	Analog inputs/outputs		Yes	yes
B.34	Digital inputs/relay outputs		Yes	yes
B.35	Fieldbus connectivity		Yes	Bop scope
B.36	SCADA Integration with NSCC		Yes	yes
	Product Compliances and Standards			

B.37	Applicable Standard (latest)		IEEE 1547-2018, IEEE 2800-2022, UL 1741-SB, IEC 62477- 1, IEC 62109-1/2, IEC TS 62786-3, IEC 61000-6-2/6-4, IEC 62116	IEC 62477, IEC61000, IEC 62909-1, IEC 62509, IEC 62109, IEC62271-202, EN 50549, IEC 50530, IEEE 519, G99 etc.
	Grid support and grid functions			
B.38	Low Voltage Ride Through		Yes	Yes
B.39	Reactive Power Compensation		Yes	Yes
B.40	Grid Forming Capability		Yes	Yes
C	Power Plant Controller			
C.1	Model			EnOS PPC
C.2	Manufacturer			Envision
C.3	Country of Manufacture			China
C.4	Dimensions			
	Length	mm		440
	Width	mm		280
	height	mm		88
D	Grid Code Requirements as per the clauses in Functional Performance Requirements, Grid Connection code and embedded generation code (Annex A ,B and C of Volume I)			
D.1	NSSC Communication Provisions		Yes	Yes
D.2	Backup Communication Provisions		Yes	Yes
D.3	Inverter protection Requirements		Yes	Yes
D.4	Backup protection Requirements		Yes	Yes
D.5	Frequency Response Requirements		Yes	Yes
D.6	Voltage control requirement		Yes	Yes
D.7	Power Factor and reactive power support		Yes	Yes
D.8	NVD protection		Yes	Yes
D.9	Low voltage ride through (LVRT)		Yes	Yes
D.10	Harmonics Compliance		Yes	Yes
D.11	Flicker Compliance		Yes	Yes
E	Grid Code Compliance Meter as per the clause A.05.22 in Functional Performance Requirements			
E.1	Real-time electrical parameter measurement			Meter is Bop scope
E.2	Power quality monitoring			Meter is Bop scope

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E.3	Fault ride-through monitoring			Meter is Bop scope
E.4	Communication and control integration			Meter is Bop scope
E.5	Data storage and reporting			Meter is Bop scope